NANGILI CHANDRA SEKHAR

SDM Deployment Engineer

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LinkedIn | Portfolio

PROFESSIONAL SUMMARY

SDM Deployment Engineer with 2.5+ years of experience deploying, integrating, commissioning and validating Nokia SDM and 5G core network functions for Rakuten Mobile through Tech Mahindra. Hands-on with Nokia HSS, HLR, UDM, AUSF, SDL, EIR, One-NDS and Oracle PCRF across LTE, VoLTE, IMS and 5G Core (SBA), deploying containerised network functions with Helm charts and Kubernetes. Skilled in subscriber data schema and authentication parameter configuration, database synchronisation and migration, Diameter and SIP interface validation, SIT/UAT execution, and end-to-end LTE/VoLTE/5G service verification. Strong in Red Hat/CentOS Linux, Bash scripting and kubectl-level validation, with MOP, LLD/HLD inputs, test and deployment reporting.

TECHNICAL SKILLS

SDM & Nokia Platforms: SDM, Nokia HSS, Nokia NT HLR, Nokia UDM, Nokia AUSF, Nokia SDL, Nokia EIR, One-NDS, Oracle PCRF, Ribbon, Nokia/Mavenir CFX, Nokia CSF / CPC / PCM concepts

5G Core & Telecom: 5G Core Service-Based Architecture (SBA), AMF, UDM, AUSF, NRF, SMF, SMSF, UPF, PCF, LTE EPC, NSA/SA architecture, VoLTE, IMS (P-CSCF, I-CSCF, S-CSCF, TAS), HLR, MME, SGW, PGW, DRA, STP, SBC, DNS, subscriber provisioning, mobility management, authentication and authorisation

Protocols: Diameter (S6a, S6d, Cx, Sh, Gx, Rx), SIP, GTP, S1AP, LDAP, HTTP/2, DNS, SCTP, TLS/SSL, PCAP capture and protocol trace analysis

Cloud & Linux: Kubernetes, Helm charts, kubectl, Docker containers, CNF/VNF, cloud-native deployment, OpenShift, VMware, OpenStack, AWS, Keycloak (user/group and secret provisioning), Linux (Red Hat/CentOS), Bash shell scripting, SSH

Databases: Cassandra, MySQL, Oracle, NoSQL, database replication and backup, subscriber data consistency checks

Tools: Wireshark, NAM, RCP Symworld, WinSCP, Xshell, PuTTY, MobaXterm, BI Studio, Python, Microsoft Office (Word, Excel, PowerPoint)

PROFESSIONAL EXPERIENCE

Tech Mahindra Ltd. | Network Engineer - Telecom Core / SDM Deployment June 2024 - Present

Client: Rakuten Mobile

SDM & 5G Core Deployment, Integration & Commissioning

- Deploy, configure and commission Nokia SDM network elements (HSS, HLR, UDM, AUSF, SDL, EIR) end to end, from installation and node configuration through service activation and deployment sign-off.
- Deploy 5G UDM cloud-native network functions using Helm charts on Kubernetes - chart configuration, installation, upgrade and validation of pods, services and application status with kubectl.
- Integrate SDM with HSS/UDM/AUSF, PCRF/PCF, MME, AMF, NRF and IMS core elements (P-CSCF, S-CSCF, I-CSCF, TAS), validating Diameter and SIP interfaces end to end after every integration.
- Configure subscriber data schemas, service policies and authentication parameters for LTE, VoLTE and 5G subscriber profiles, and create users/groups in Keycloak for secret provisioning.
- Perform subscriber database synchronisation and data migration between Kubernetes-hosted SDM pods, verifying record counts, replication status and data integrity before service release.

Deployment Testing & Validation

- Execute SIT and UAT for SDM deployments, covering subscriber provisioning, authentication, registration, mobility and roaming scenarios, and record results in test and acceptance reports.
- Validate Diameter interfaces and subscriber provisioning flows using protocol traces, transaction logs and PCAP captures to confirm correct request/response behaviour before sign-off.
- Validate 5G registration, authentication, PDU session establishment and deregistration, plus LTE data and VoLTE call setup, end to end after each deployment and upgrade.
- Perform ATP, pre-deployment checks and post-deployment service validation to confirm node health, service status and KPI compliance at handover.

Upgrades, Migration & Deployment Documentation

- Execute software upgrades, patch and Helm chart deployments for Nokia/Mavenir CFX alongside vendor and L2/L3 teams, including pre-upgrade backup, rollback planning and post-upgrade validation.
- Prepare MOP/RMOP, LLD/HLD inputs, test reports, deployment reports and handover documentation, and coordinate with vendors and L2/L3 teams during deployment windows.

EDUCATION

Sri Venkateshwara Engineering College 2019 - 2023

Bachelor of Technology (B.Tech), Electronics and Communication Engineering | CGPA: 8.22 / 10

LANGUAGES

English, Telugu, Hindi, Kannada